

Bilkent University

EEE 313

Charge Pump - Design Phase



I affirm that I have not given or received any unauthorized help on this report
and that this work is my own.

1. Introduction

The objective of this project is to design a low cost charge pump circuit with a single input power source of 6V DC, satisfying all requirements like output load voltage, efficiency, voltage regulation, and cost - effectiveness.

To meet these goals, the circuit utilizes an NE555 timer configured in astable mode to generate a switching signal. This signal is buffered by a BJT push-pull driver (using BC238 and BC308) to provide sufficient current to a multi-stage diode - capacitor voltage multiplier network. LTspice was used to build and test the circuit.

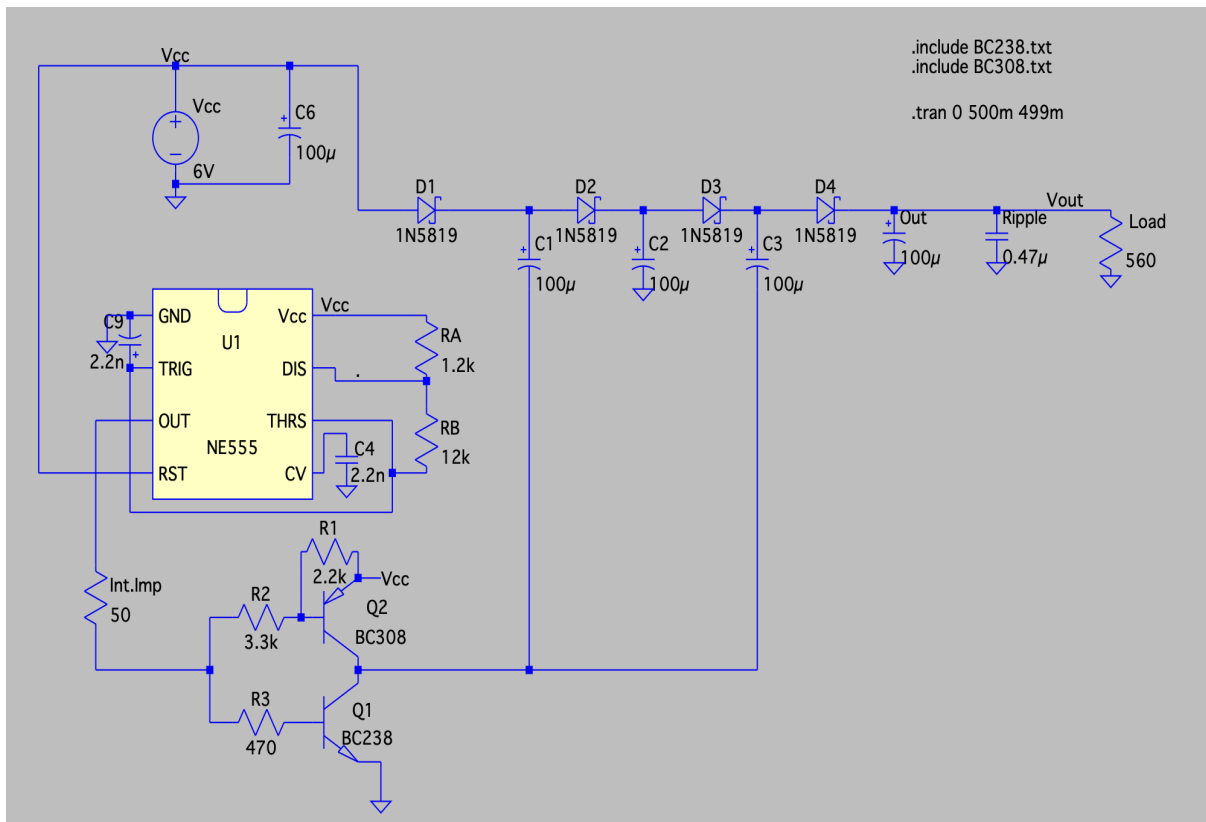


Figure 1. LTspice circuit design

2.1 NE555 Configuration

The timing characteristics are determined by resistors RA, RB, and the ceramic capacitor C9. To achieve high performance the following design choices were made:

1. **Astable Mode:** The NE555 is configured to oscillate continuously using a 6V power supply.
2. **50% Duty Cycle:** By choosing values for RA and RB (1k and 12k respectively), the duty cycle is $\frac{RB}{RA+2RB} = 47.6\%$, which is acceptably close to the ideal value..
3. **Frequency:** The formula for frequency is $F = \frac{1.44}{(RA+RB)C9} = 25.974\text{kHz}$.

2.2 Driver Configuration

BJT push-pull driver is implemented to act as an inverter and a high-current buffer. This stage is critical for ensuring that the charge pump capacitors C1 and C3 can be charged and discharged rapidly without overloading the our NE555 timer.

The driver utilizes a complementary pair of BJTs: an NPN BC238 (Q1) and a PNP BC308 (Q2). They are configured as a common-emitter inverter: When the NE555 output is high, Q1 is forward-biased and enters saturation, pulling the output node to GND. Conversely, when the NE555 output is low, our Q2 is forward-biased so it pulls the output node to the power supply Vcc voltage.

While the NE555 can typically provide up to 200mA, driving a heavy load like 560 Ohm and having two pumping stages can cause heating and voltage drop. This BJT stage provides a low - impedance path to the supply rails, with sharp switching transitions.

2.3 Charge Pump and Output Stage

To provide the necessary >15V to the output, we provided a two stage voltage multiplier network with 4 Schottky diodes (D1-D4) and three electrolyte capacitors 100uF each (C1-C3). We have two pumping capacitors, so the voltage stacks twice, first C1 is charged with Vcc + driver output (approximately 6V as well), after it translates to the C2 as a reservoir, after in C3 the voltage stacks again with driver out. Diodes are used to prevent the reverse current flowing. Moreover, diodes have internal voltage drop around 0.3V (Vd). Consequently, we have Vcc + 2x driver output (with some error) - 4Vd (diode voltage drop). After all of this voltage is stored in the final capacitor Cout with our load and the Cripple which is used to prevent high frequency switching noise.

3.0 Specifications

We have 4 criteria to meet to satisfy the project requirements.

3.1 The Average Output Voltage >15V

Our $\tau = R_{load} * C_{out} = 56ms \Rightarrow$ steady state around 280ms, we use .tran 500m 499m to verify the values.

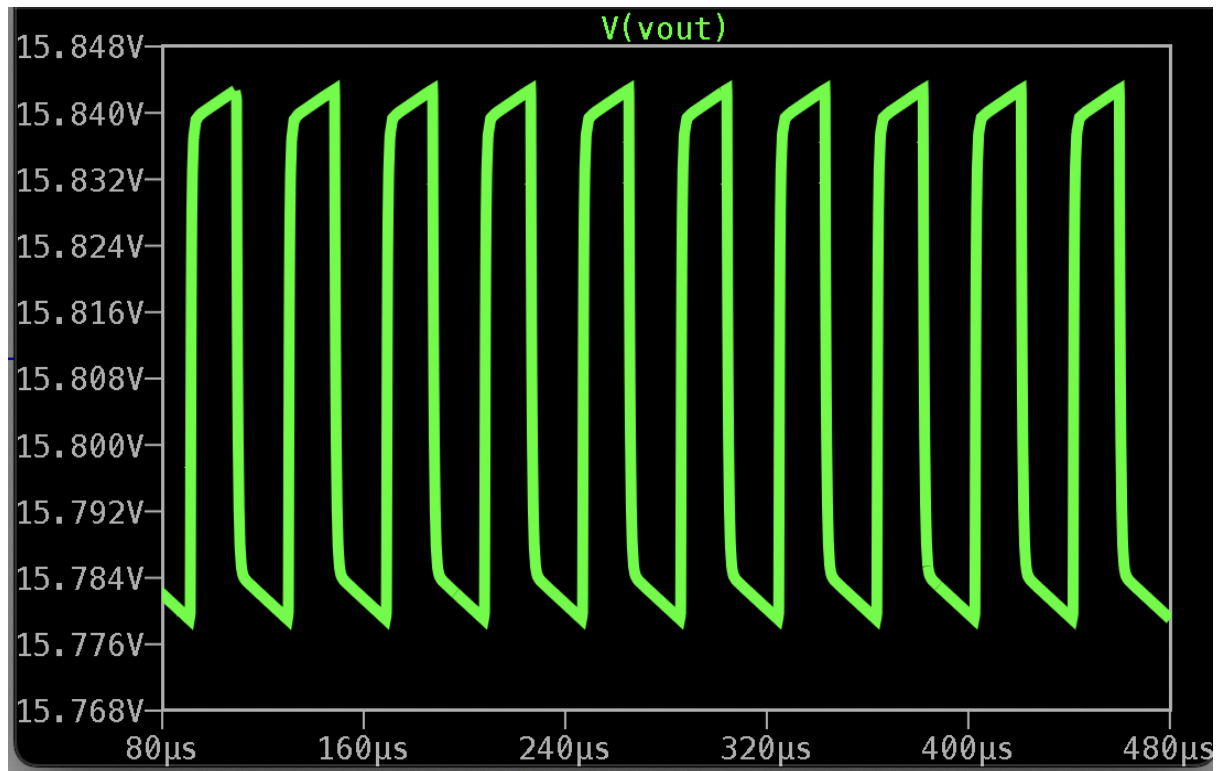


Figure 2. Vout

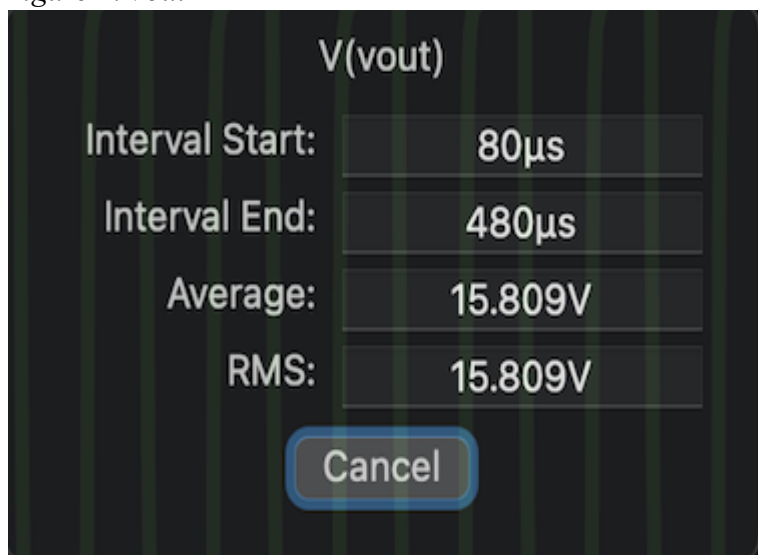


Figure 3. Average Vout

As we can see our averaged Vout is **15.809** which is >15V, satisfying the first criteria.

3.2 No Load Voltage < 20V

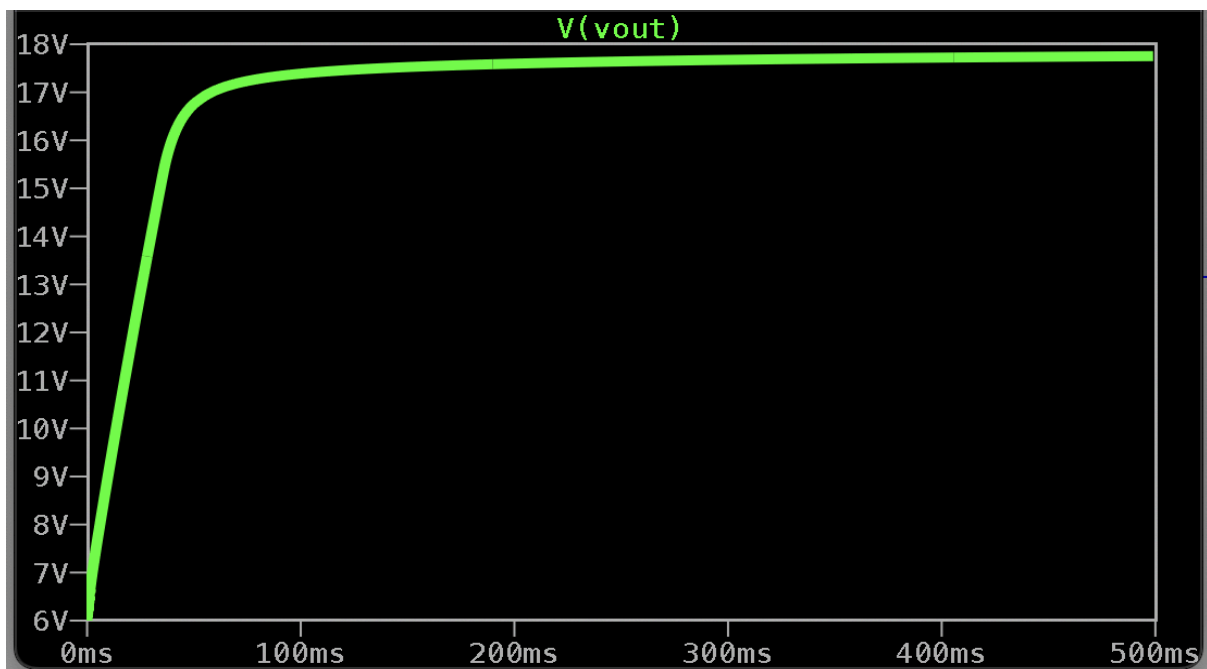


Figure 4. V_{out} with no load

Theoretically, from the formula we got in 2.3 section, **our voltage can not exceed 18V**, using only a 2 stage voltage multiplier in our circuit. Figure 4 solidifies our expectations and satisfies the second criteria.

3.3 Efficiency > 60%

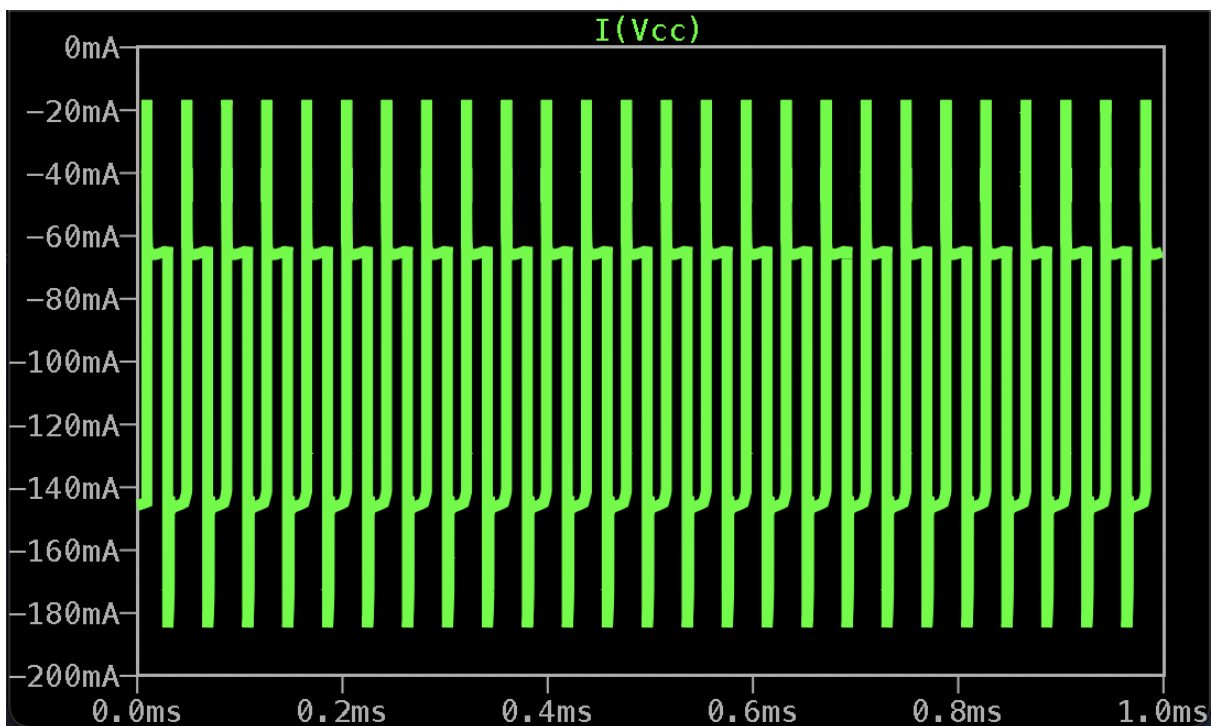


Figure 5. Power Supply Current

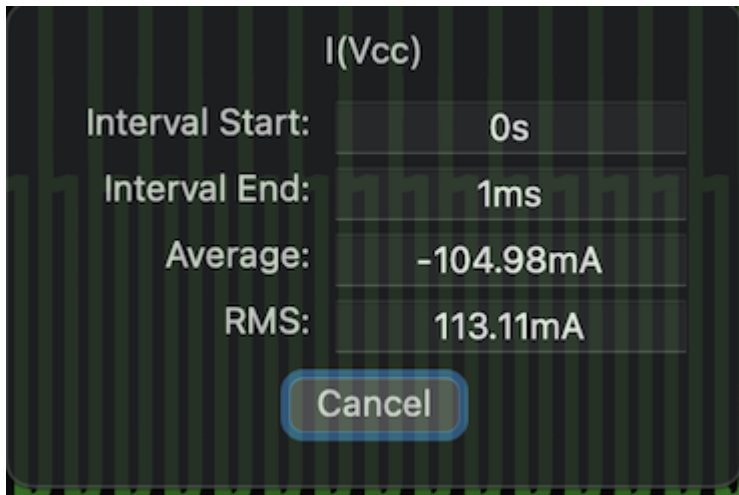


Figure 6. Averaged Supply Current

With an averaged current power supply, we have an efficiency formula as:

$$\eta = \frac{V_{out}^2}{R_L * V_{cc} * I_s} = 81.54\% > 60\%.$$

3.4 The Ripple Output Voltage

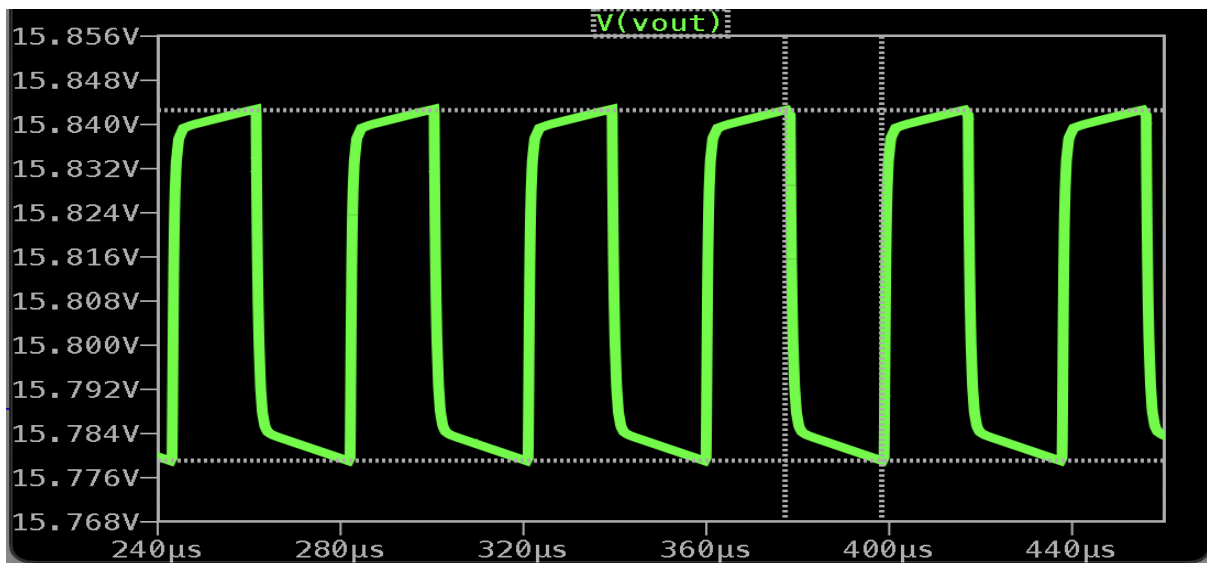


Figure 7. Vout ripple voltage

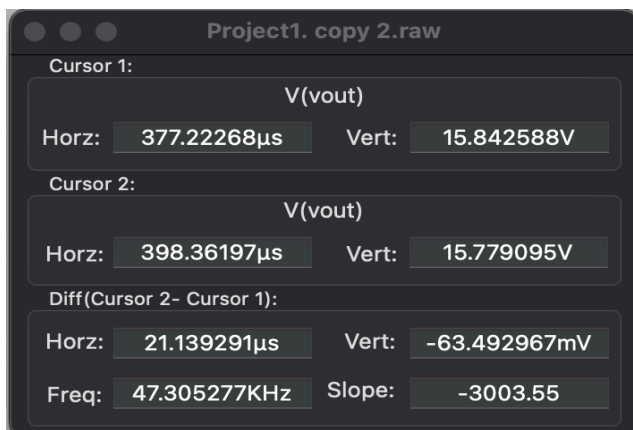


Figure 8. *V_{out} ripple voltage difference*

From Figure 8 our ripple voltage is **0.063V < 0.1V**.

3.5 Minimized Cost < 20TL

Component	Type	Quantity	Unit Price (TL)	Total (TL)
NE555	Timer IC	1	6.49	6.49
BC238	NPN BJT	1	1	1
BC308	PNP BJT	1	1.07	1.07
1N5819	Schottky Diode	4	1.06	4.24
100μF 25V	Electrolytic Capacitor	5	0.7	3.5
2.2nF	Ceramic Capacitor	1	0.29	0.29
0.47μF 63V	Ceramic Capacitor	1	0.38	0.38
1/4W Resistors	Standard Resistors	6	0.12	0.72
Grand Total				17.69 TL

Table 1. *Cost Table*

Our total cost is calculated to be **17.57TL < 20TL**.

4.0 Conclusion

Our circuit successfully and comfortably achieved the required criteria, achieving more than 80% efficiency, and almost 16V output. At the same time, the rippling is moderate, and the voltage without load can not get above 18V in either case. Finally, we have spare 2.5TL left in our budget, with only 17.57TL total cost.